

Article

Hybrid Solar PV–Agro-Waste-Driven Combined Heat and Power Energy System as Feasible Energy Source for Schools in Sub-Saharan Africa

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Abstract: Poor access to electricity in rural communities has been linked to a poor educational system, as electricity is essential for supporting laboratories, technical practice, and long study hours for students. Therefore, this work presents the techno-economic analysis of a hybrid solar PV–agro-wastes (syngas) energy system for electricity, heat, and cooling generation to improve energy access in rural schools. The system is located in Ghana at Tuna (lat. 9°29'18.28" N and long. 2°25'51.02" W) and serves a secondary school for enhanced quality education. The system relies on agro-waste (gasifier-generator) and sunlight (solar PV), with a battery energy storage system, to meet the school's energy demand. The study employs HOMER Pro Version 3.16.2 software to comprehensively analyze technical, economic, and environmental aspects. The system can generate 221,621 kWh of electricity (at a unit cost of electricity of 0.295 EUR/kWh) and 110,896 kWh of thermal energy yearly. The cost of electricity from the proposed system is cheaper than the cost of electricity from an equivalent diesel generator at 0.380 EUR/kWh. The thermal energy can meet the heating demand of the school in addition to powering a vapor absorption chiller. The system is environmentally friendly, with the capacity to sink 0.526 kg of CO₂ yearly. Government policies that moderate interest rates for bioenergy/solar PV systems and social solution on feedstock pricing will favor the economic sustainability of the proposed system. The system will address the energy access challenge (SDG 7), enhance the quality of education (SDG 4), and contribute to climate mitigation through carbon sequestration (SDG 13).



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1. Introduction

Access to reliable and affordable energy (SDG7) is critical to global socioeconomic growth. However, for a sizable segment of the world's population, energy access challenges persist, inhibiting adequate access to modern energy services. The United Nations estimates that 13% of the world's population lacks access to electricity, while 40% lacks access to clean cooking services [1]. Inadequate energy access hampers progress in economic and human development, particularly in low- and middle-income nations (LMIN) [2]. The current pace of advancement towards sustainable energy goals is falling short of achieving United Nations Sustainable Development Goal 7 by 2030, which aims to ensure access to clean and affordable energy. Energy access is crucial for various sectors, including education, agriculture, business, communication, healthcare, and transportation [3]. Ghana's

electricity production is largely skewed towards traditional sources, accounting for around 68.8% of the total energy mix [4]. The country has grappled with inadequate electricity supply challenges [5], resulting in an average daily production loss of approximately USD 2.1 million over the last decade [6]. The most recent data on Ghana's electrification rate indicate that the country's electrification rate is close to 86%, as shown by World Bank data in Figure 1. Looking at the increase in electricity access in recent years, it can be seen that Ghana has managed to maintain growth in electricity access of approximately 20% every 10 years. This same behavior can be seen in other African countries, to a greater or lesser magnitude, but with a steady increase. Figure 1 shows the trends in some countries on the African continent.

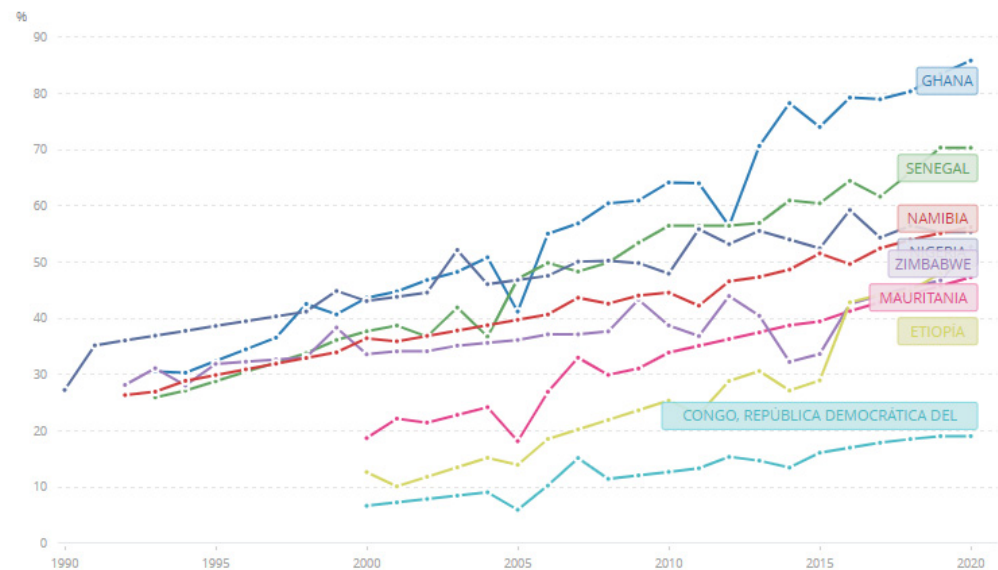


Figure 1. Access to electricity in African countries since 1990 (Source: World Bank).

Despite Ghana's relatively high electricity access rate, power disruptions and energy poverty remain significant issues, especially in rural communities [7,8]; these issues may be attributed to the disparities in educational outcomes between urban and rural areas [9]. Ongoing efforts are being made to address these challenges and further improve the education system. Addressing this requires strategies such as expanding the generation mix to include diverse renewable energy sources in addition to traditional ones [10,11]. Energy's significance in modern education lies in its role in fostering suitable learning environments and enabling creative teaching approaches, and the absence of access to modern energy services at home can detrimentally affect the quality of schooling and education [12].

There are some case studies of Ghanaian schools facing energy shortages in the literature, such as the study of Nduhuura, Garschagen [8], which highlights the impact of electricity outages on urban households in Ghana, including the impact on school-going children; the Ahali [13] study, which discusses the challenges confronting Ghana's energy sector and its impact on education; and the assessment by Cobbold, Owusu [14] of the electricity sector's effects on education. These studies, amongst others, collectively underscore the presence of energy challenges in Ghana, notably in secondary schools, where energy shortages can negatively affect students' performance and teachers' effectiveness [9].

In educational settings like secondary schools, energy poverty leads to various issues, such as a decrease in teaching resources and classroom materials, limited student access to essential assignment resources, heightened challenges for teachers in reproducing school materials and using the internet for research, and discouragement to teachers from working in areas with energy access challenges. Energy poverty causes added complexity for staff and school administration in tasks like record-keeping, grading, and attendance tracking, and necessitates manual paper-based methods, which are both inefficient and wasteful [15,16]. Adequate electricity for modern media and tools enhances student performance and brings added benefits like boosting attendance, retention, and achievements for both students and teachers [17].

Renewable energy solutions can assist in addressing electricity access challenges in Ghanaian schools [10,18]. Ghana has significant solar energy potential due to its equatorial location, making it suitable for both concentrating and non-concentrating solar technologies. The abundant solar resource holds the promise of enhancing the energy landscape, ensuring reliable and high-quality energy services for all schools in Ghana [19].

One such renewable solution can be found in the adoption of solar photovoltaic systems [20,21]. Solar photovoltaic (PV) systems harness sunlight to generate electricity for various applications, including schools. PV cells directly convert sunlight into electricity, and their efficiency has greatly improved. These systems can be organized into arrays of varying sizes, catering for needs such as powering water pumps, supplying energy to homes, or contributing to larger utility-scale electricity generation [22,23].

Energy challenges, including poverty and insufficient electricity supply, are prevalent in Ghana's agrarian communities. These challenges not only affect education but also hinder agricultural productivity due to the limitations on the utilization of modern energy services [13,24,25]. Furthermore, another potential renewable energy solution can be sourced from agro-waste [26,27]. In Ghana, agro-wastes like crop residue and animal byproducts hold the potential for energy production [28,29]. Employing bioenergy technology, these agro-wastes can be transformed into syngas and biogas, serving purposes like cooking, heating, and generating electricity. Additionally, these materials can be utilized to create biofuels, including ethanol and biodiesel [30–32].

Enhancing the energy potential of renewable solutions leads to developing a reliable energy supply using hybrid systems that combine multiple renewable sources [33]. The affordability and accessibility of solar energy have risen in Ghana, largely due to a notable reduction in solar panel costs and related technologies [34]; and additionally, the secondary school's location in a primarily agrarian region guarantees a continuous biomass feedstock supply. Furthermore, a hybrid system merging solar and agro-waste (syngas) technologies will offer reliable power generation, overcoming limitations due to weather conditions or system availability.

Based on the premise above, this article aims to assess the feasibility of a solar PV–agro-waste (syngas) hybrid heat and power energy system for a Ghanaian secondary school to address energy deficits and enhance quality education. The study employs HOMER software [35] for comprehensive analysis, which encompasses technical, economic, and environmental aspects, providing a holistic assessment of the proposed solution's viability. The innovative aspect of the proposed system lies in its ability to leverage the advantages of two renewable energy sources—solar and biomass—to generate electricity, heat, and cooling simultaneously. Most renewable energy systems typically focus on either electricity or heat generation alone. In rural schools, there is a critical need for electricity, hot water for sanitation, and cooling for food storage or comfort, especially in hot climates. However, few studies have integrated all these energy requirements into a single system. This comprehensive approach addresses diverse energy needs cost-effectively, which can enhance the operational efficiency of schools in sub-Saharan Africa.

This article begins with an introduction to set the context of the paper, followed by the methodology section, which presents and explains the system description, HOMER software application, and the parameters considered for the analysis. The results and discussion section presents the findings, accompanied by relevant discussions, and the article concludes with a summary of implications, recommendations, and the contribution of the hybrid system to sustainable energy access and education

2. Problem Formulation and Solution Methods

The hybrid solar PV–biomass (syngas) (HSB) system was modeled and simulated by considering relevant thermodynamic and economic parameters. First, the location for the demonstration of the system is described, followed by the description of the proposed system. The thermodynamics and economics models of the hybrid system immediately follow. Figure 2 conceptualizes the methodological flow and the relationship between relevant input parameters and the corresponding output parameters of the problem under investigation.

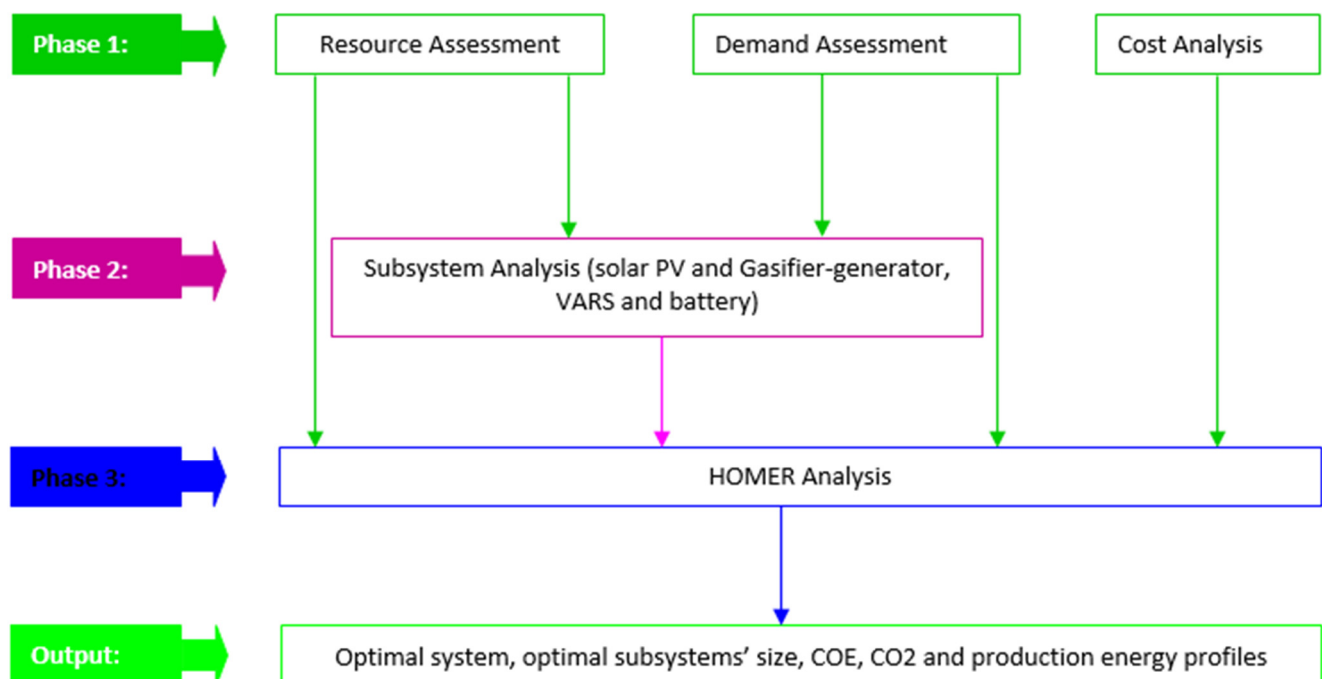


Figure 2. Input and output flow block diagram.

2.1. Study Location

The proposed location of the demonstrated system is Tuna Senior High/Technical School Education Centre (TUSEC) located in Tuna, Ghana, (lat. $9^{\circ}29'18.28''$ N and long. $2^{\circ}25'51.02''$ W), shown in Figure 3 with a red patch. It is a hot and dry location in fairly windy conditions. Tuna is the second most populous community in the Sawla-Tuna Kalba District in the Savannah Region, Ghana. It is about 26 km North of Sawla, just west of the N12 highway. TUSEC currently consists of 22 buildings with a capacity of about 1000 students.

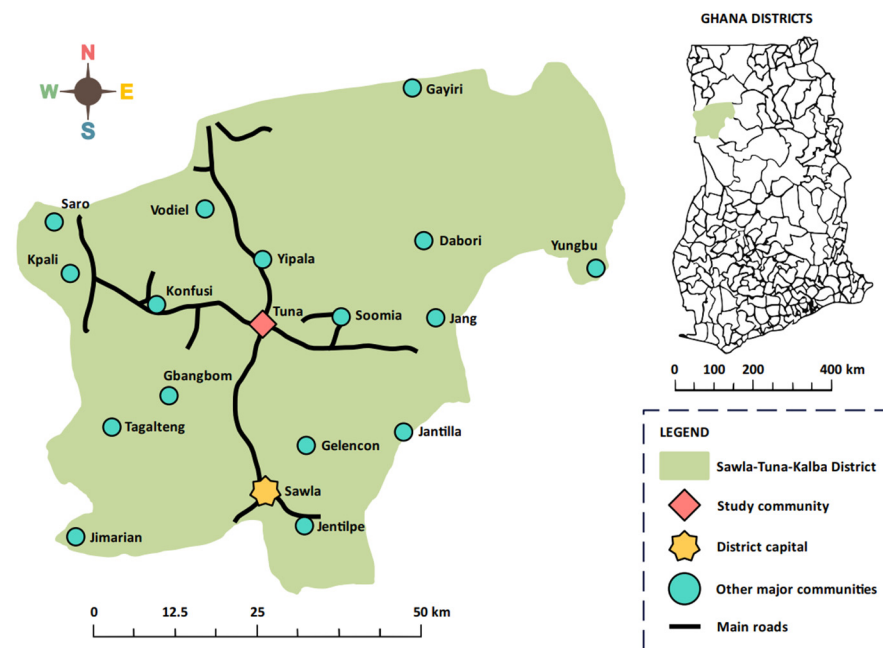


Figure 3. Geographical location of Tuna community (Ghana).

2.2. System Description

As shown in Figure 4, agro-waste (groundnut shells) enters the gasification unit (gasifier), which converts the agro-waste to synthetic gas (syngas). The production process of syngas involves thermochemical reactions (pyrolysis, oxidation, and reduction) that convert raw materials into a gas mixture in an oxygen-limited environment. The process typically occurs at high temperatures ranging from 800 °C to 1200 °C. After gasification, the raw syngas is cleaned to remove impurities, including tar, particulate matter, sulfur compounds, and nitrogen compounds such as ammonia (NH₃). Syngas is composed of hydrogen (H₂), carbon monoxide (CO), and carbon dioxide (CO₂), along with smaller amounts of methane (CH₄), nitrogen (N₂), and other trace gases. On leaving the gasifier unit, the syngas enters the gas engine (NPT 20GFT gas genset). The gas engine converts the syngas to electricity (alternating current) and thermal energy. The thermal energy, which is derived by a waste heat recovery system (from the flue gases at 130 °C), is partly used to meet the heating demand of the facility (TUSEC) and to power a vapor absorption refrigeration system to meet the cooling needs of the facility at 15 °C. A solar photovoltaic (PV) system converts the solar radiation to electricity (direct current), which is then converted to alternating current (AC) by the hybrid converter unit (controller and inverter). The excess electricity generation from the gas engine and solar PV is stored in the battery as a direct current, which is later released as alternating current to the facility during capacity shortages through the hybrid converter. The communication between the facility (the demand side), converter, and battery is coordinated by a communication signal (the broken purple line) embedded in the controller. The continuous green, purple, red, and blue lines show the flow of resources, electricity, hot fluid, and cooling, respectively. The detailed analysis of the VARS is outside the scope of the current analysis.

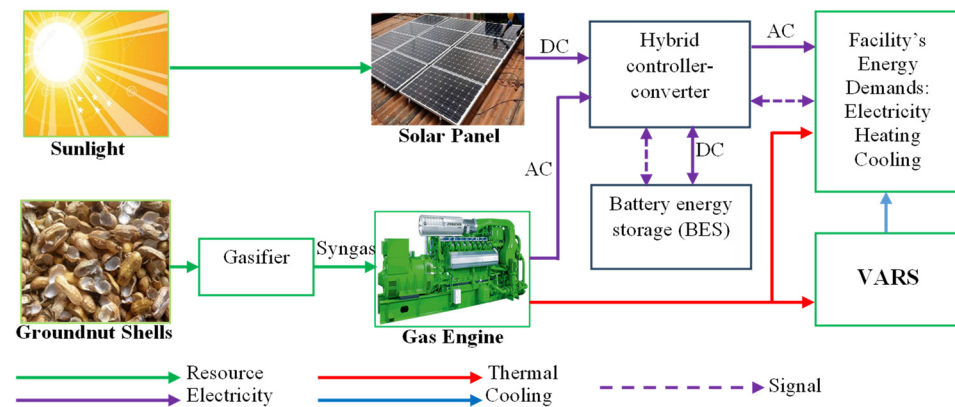


Figure 4. Process diagram of the system.

2.3. Solution Formulation

2.3.1. Solar PV Power

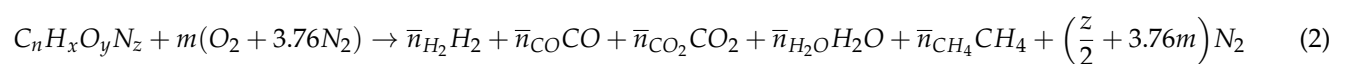
The solar PV system generates direct current (DC) voltage when solar irradiance is incident on the PV arrays, and the power output from the PV can be computed as follows [35,36]:

$$P_{out} = P_{rated} \times f_{pv} \left(\frac{G}{G_{ref}} \right) \times \left[1 + K_{T,pv} (T_c - T_{ref}) \right] \quad (1)$$

where P_{rated} (kW), f_{pv} (%), G_{ref} (kW/m²), G (kW/m²), $K_{T,pv}$, T_{ref} and T_c are the PV-rated power at standard test condition (STC), the PV derating factor, the radiation at STC, the global solar irradiance incident on the solar PV surface, the temperature coefficient of the solar PV module, the cell temperature at STC, and PV cell temperature, which can be approximated as $T_c = T_{amb} + 0.0256G$ according to Duffie and Beckman [37], respectively; T_{amb} is the ambient temperature. The effects of wind and humidity on solar PV performance were not considered since studies have shown that they have moderate effects on the PV module efficiency [38,39].

2.3.2. Gasifier

To utilize the energy trapped in the agro-waste, a gasifier that converts the agro-waste into syngas is modeled according to Arafat and Jijakli [40] and Allesina, Pedrazzi [41].



where $(C_nH_xO_yN_z)$ is the chemical formula obtained by the method of Chandrappa and Das [42]; $\bar{n}_i$ is the molar composition of the i th component of the syngas obtained from the following atomic balance, chemical equilibrium reactions, and energy balance:

$$\text{Carbon balance : } n = \bar{n}_{CO} + \bar{n}_{CO_2} + \bar{n}_{CH_4} \quad (3)$$

$$\text{Hydrogen balance : } x = 2\bar{n}_{H_2} + 2\bar{n}_{H_2O} + 4\bar{n}_{CH_4} \quad (4)$$

$$\text{Oxygen balance : } 2m + y = \bar{n}_{CO} + 2\bar{n}_{CO_2} + \bar{n}_{H_2O} \quad (5)$$

The equilibrium reactions considered as taking place in the gasifier are the water gas shift and methane reaction, as demonstrated by other studies, e.g., Houshfar, Becidan [43] and Jarungthammachote and Dutta [44]. These are mathematically represented in Equations (6)–(9):

$$\text{Water gas shift : } K_{WGS}(\bar{n}_{CO})(\bar{n}_{H_2O}) = (\bar{n}_{CO_2})(\bar{n}_{H_2}) \quad (6)$$

$$\text{Methane reaction: } K_{MR}(\bar{n}_{H_2})^2 = (\bar{n}_{CH_4})(\bar{n}_{tot}) \quad (7)$$

$$\bar{n}_{tot} = \bar{n}_{H_2} + \bar{n}_{H_2O} + \bar{n}_{CH_4} + \bar{n}_{CO} + \bar{n}_{CO_2} + \left(\frac{z}{2} + 3.76m\right) \quad (8)$$

$$\ln(K_i) = -\frac{\Delta \bar{G}_{T,i}^0}{RT} \quad (9)$$

Details of these reactions can be found in Houshfar, Becidan [43] and Jarungthamchote and Dutta [44]

Gas Engine

The fuel consumption (f_{con}) of the gas engine can be computed according to Lambert, Gilman [35] using Equation (10):

$$f_{con} = f_o Y_{gen} + f_1 P_{gen} \quad (10)$$

where f_{con} (kg/h) is the fuel consumption rate, f_o (kg/h/kWrated) is the generator fuel curve intercept coefficient, f_1 (kg/h/kWoutput) is the generator fuel curve slope, Y_{gen} (kW) is the rated capacity of the generator, and P_{gen} (kW) is output of the generator matched with demand profile.

2.3.3. Vapor Absorption Refrigeration System (VARs)

The coefficient of performance (COP) of the vapor absorption system can be approximately (neglecting the energy demand of the pump) estimated using Equation (11):

$$COP = \frac{\dot{Q}_{eva}}{\dot{Q}_{gen}} \quad (11)$$

where $\dot{Q}_{eva}$ (kW) and $\dot{Q}_{gen}$ (kW) are the derived cooling capacity and thermal load of the generator; 0.7 (–) was adopted as the COP for the current study based on the adopted VARs. The VARs' generator and evaporator temperatures will operate at 120 °C and 15 °C, respectively.

2.3.4. HOMER Analysis

HOMER software (HOMER Pro Version 3.16.2) is a high-fidelity mini-grid optimization and simulation computer package that guides techno-economic decision-making. HOMER software is a widely used tool for conducting life cycle cost analysis (LCCA) for energy systems. It is specifically designed to model and optimize hybrid renewable energy systems, including combinations of solar PV, wind, diesel generators, battery storage, and other resources. The life cycle cost analysis performed by HOMER helps identify the most cost-effective configuration of an energy system throughout its operational lifetime, considering various economic, technical, and environmental factors [45].

Figure 5 shows the configuration of the proposed system implemented in the HOMER software, showing the system's geographical location and various components. To assess the feasibility and the optimal design of the hybrid system, the HOMER software is used for the modeling. The optimization computational algorithms of the HOMER software allow rapid and robust techno-economic evaluations of various energy technology options by accounting for the cost of energy alternatives and the availability of renewable energy resources. HOMER uses the load demand (the electricity demand profile), the resources (solar radiation, groundnut shell, and temperature variation), the components' technical characteristics, the components' cost details, the constraints, the system control, and the emission data as inputs to simulate various feasible configurations, and ranks them by the net present value (NPV) over the project lifetime (25 years in this case). The NPV, the present cost of the system minus the sum of revenues, serves as the objective function.

HOMER obtains the best system configuration after balancing energy demand and supply for each hour of the system simulation [46].

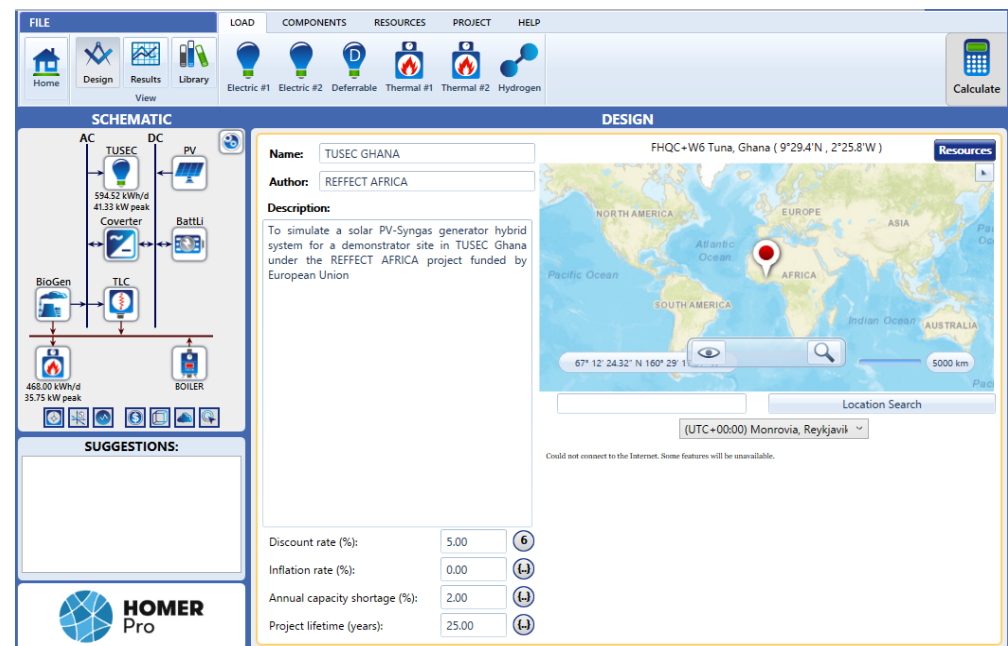


Figure 5. The hybrid system on HOMER's schematic window.

The NPV of the system can be related to the annualized life cycle cost (ALCC) of the system, which represents the present-day worth of money, as presented in Diemuodeke, Addo [45]:

$$ALCC = F(i, N)NPV \quad (12)$$

where $F(i, N)$ is the system capital recovery factor, which is related as:

$$F(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (13)$$

The levelized cost of electricity (LCOE), which represents the average cost per kWh of the electrical energy generated by the system, can be calculated as:

$$LCOE = \frac{ALCC}{E_s} \quad (14)$$

where E_s (kWh/year) is the actual electrical energy supplied to the facility by the system.

All the basic technical and economic calculations, i.e., Equations (1)–(14), are considered in the solution method, with some embedded in the HOMER software computational algorithm. The system was modeled as a combined heat and power (CHP) system; see Figure 5. The system comprises solar PV (PV), a gas generator (BioGen), a battery (BattLi) and a converter/controller (Converter), with electrical (TUSEC) and thermal loads. The electrical load serves the facility, and excess electrical energy is used to serve the dumped load as a resistive heater, modeled as a thermal load controller (TLC) in HOMER. The CHP system was implemented by incorporating the thermal load and boiler to serve the facility and the vapor absorption refrigeration system (VARS). The combined heat and power module enables users to model a gas generator as a supplier of electricity and heat. The module consists of two thermal loads, the boiler component, the thermal load controller component, and the heat recovery ratio parameter. The heat recovery ratio enables the balancing of the resources (the syngas in the present case) to balance the electricity and heat needs of the facility.

3. Energy Demand and Resource Assessments

3.1. Energy Demands

Figure 6 shows the daily electrical energy demand and profile for the facility at TUSEC. The daily electrical energy demand was obtained based on records of the daily energy consumption of the facility and complemented by an energy audit (which involves interviews of staff and students) to ascertain the pattern of appliances operating within the facility. Thermal energy demand for the facility was not readily available. However, the thermal load of 11 kW of the vapor absorption refrigeration system was incorporated, and a provision for excess thermal energy production was made to take care of the facility’s thermal load.

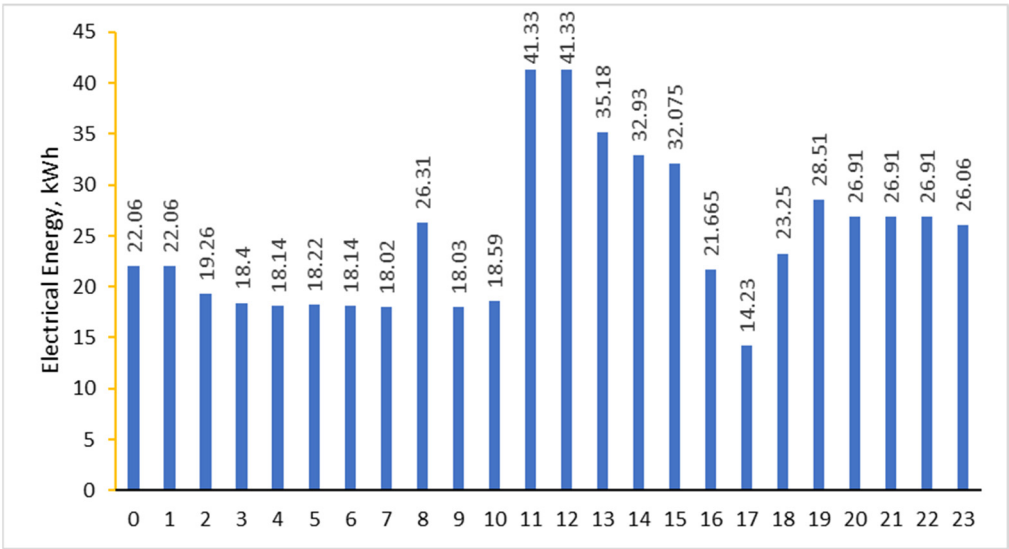


Figure 6. Daily electricity demand profile.

3.2. Resource Assessments

The solar and biomass resources used for the optimization and simulation are presented in Table 1. The solar resource was based on NASA’s monthly average solar Global Horizontal Irradiance for Tuna (9°29.4’ N, 2°25.8’ W) embedded in the HOMER database, which has been demonstrated to be accurate within Ghana [47]. The biomass resource was derived from the yearly technical potential for groundnut shells, which is assumed to be uniform throughout the year on the ground that residues will be stored for the off-season period. The annual production of shelled peanuts facilitates estimating the corresponding annual residue output from groundnuts through a parameter known as the residue-to-product ratio (RPR). This parameter establishes the relationship between the mass of specific residues generated by the crop and the actual product derived from the crop. According to Kemausuor et al. [48], the value of 0.35 was adopted for the RPR. Biomass consumption for the 20 kWe gasification plant was about 25–30 kg/h, while the syngas generation was around 70 kg/h (60–65 Nm³/h), which gives a gasification ratio of 2.33 kg/kg, with an LHV value of about 4.5–5.0 MJ/Nm³ (3.88–4.31 MJ/kg). The low LHV is attributed to the significant amount of nitrogen and other non-combustible gases in the syngas.

Table 1. Solar and biomass resources.

Parameter	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Solar GHI (kWh/m ² /day)	5.74	5.98	6.11	5.92	5.73	5.03	4.51	4.22	4.66	5.32	5.46	5.63
Biomass (Mton/day)	14.66	14.66	14.66	14.66	14.66	14.66	14.66	14.66	14.66	14.66	14.66	14.66

3.3. Cost Analysis

Components' costs were sourced from the public domain, with modifications to reflect the studied area, and from the database of the simulation software, as shown in Table 2. A discount rate of 7.5% was adopted, according to [49]. The gas genset's operation and maintenance (O&M) cost was derived from the combination of the groundnut shell, transportation, pretreatment, and biochar sales costs, reflecting the studied area's context sensitivity. By incorporating all relevant costs over the lifetime of the project, including operational, replacement, and decommissioning costs, HOMER can provide a comprehensive financial assessment of the hybrid energy system.

Table 2. Cost data.

Components	Initial Cost	Replacement Cost	O&M	Source
Solar PV	834.15 (EUR/kW)	834.15 (EUR/kW)	9.43 (EUR/year)	[49]
NPT 20GFT gas genset	3000 (EUR/kW)	2500 (EUR/kW)	0.085 (EUR/op. h)	Authors ^c
Battery (Li-Ion)	518.40 (EUR/kWh)	518.40 (USD/kWh)	9.43 (EUR/year)	Website ^a
Converter	622.08 (EUR/kW)	622.08 (EUR/kW)	94.30 (EUR/kW)	HOMER database and Authors ^b
TLC	188.51 (EUR/kW)	188.51 (EUR/kW)	200 (EUR/kW)	HOMER database and Authors ^b

^a <https://source.benchmarkminerals.com/article/lithium-ion-battery-cell-prices-fall-to-110-kwh-but-raw-material-risk-looms-large-2> (accessed on 13 April 2023). ^b The costs were adjusted to reflect the peculiarity of the studied area. ^c Based on the authors' market survey.

The data presented in this study are generally limited to Ghana, but because of Ghana's geographical and socioeconomic position, they are typical of sub-Saharan Africa.

4. Results and Discussion

To meet the energy demands (thermal and electricity), the HOMER simulation and optimization software presents the results in Table 3 for the hybrid solar PV–biomass (syngas) energy system for TUSEC. The simulation assumed a 40% load factor for the gas generator. The TLC was incorporated to take care of excess electricity generation. The facility's energy demand can be optimally met with 20 kW of solar PV, 24 kW of generator capacity, and 78 kWh of battery capacity. The levelized cost of electricity (COE) is estimated at 0.295 EUR/kWh, which is above the electricity cost from the municipal electricity grid at 0.100 EUR/kWh as of March 2023 [50]. However, the cost of electricity of the proposed system is lower than electricity from diesel generators operated in Ghana, which is estimated at 0.380 EUR/kWh [51] and slightly higher compared with the cost of electricity of about 0.242 EUR/kWh for a solar PV–biogas hybrid energy system presented in Odoi-Yorke, Abaase [52]. The difference between the proposed system's LCOE and the LCOE of the system presented in Odoi-Yorke, Abaase [52] could be attributed mainly to the different energy conversion technologies deployed. The high cost could be attributed to the extra cost of the heat recovery system (TLC) without evaluating the cost of thermal energy; it is assumed to be a free service. In addition, the HSB system is environmentally friendly as it serves as an emission sink (it sinks 0.526 kg of CO₂ annually), beyond the CO₂ avoidance. The implication is that if the thermal energy service and emission avoidance and capture potential are considered in the cost analysis, the HSB system may

be economically competitive with the cost of electricity from municipal generation. In addition, the proposed HSB system's CO₂ sequestration and provision of organic fertilizer (biochar) make it environmentally superior to other competing renewable systems.

Table 3. Optimized system values.

Component	Unit	Value
Solar PV	kWp	20
Gasifier-generator	kW	24
Cooling capacity	kW	7.7
Cool room temperature	°C	15
Battery	kWh	78
Thermal Load Control (TLC)	kW	1
Converter	kW	24.8
Bus voltage	V	12
Autonomy	h	2.52
Electricity production	kWh/year	221,621
Thermal production	kWh/year	110,896
Excess thermal	kWh/year	14,536
Levelized cost of energy	EUR/kWh	0.295
System CO ₂ emission	kg/year	−0.526
Control type	-	FL

4.1. Simulations

4.1.1. Electrical Energy Consumption

The facility has electrical demand of 592 kWh/day with a peak load of 41 kW. The optimized system can meet the facility's electrical load by the generation sources presented in Figure 7. On average, the gasifier-generator meets about 81% of the facility's electrical energy. Figure 8 shows the time series electrical generation of the hybrid system, which shows the variation in power generation from the sources. The figure shows the importance of battery storage in meeting the electrical energy demand of the facility throughout the year. The negative values in Figure 8 indicate that electrical power was used to charge the battery bank.

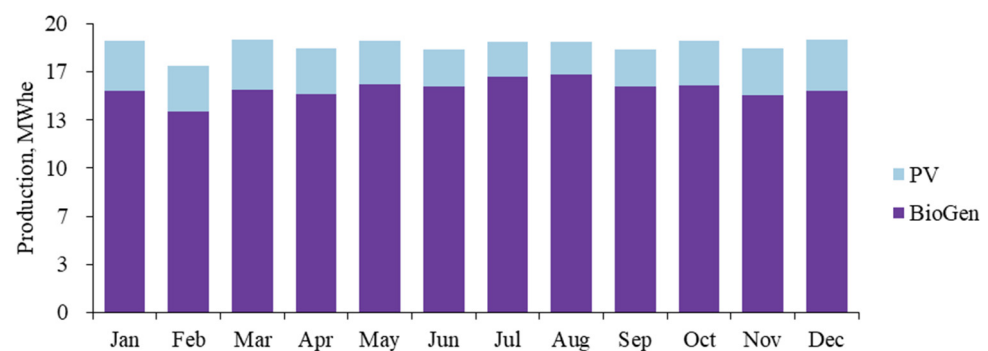


Figure 7. Electrical energy generation.

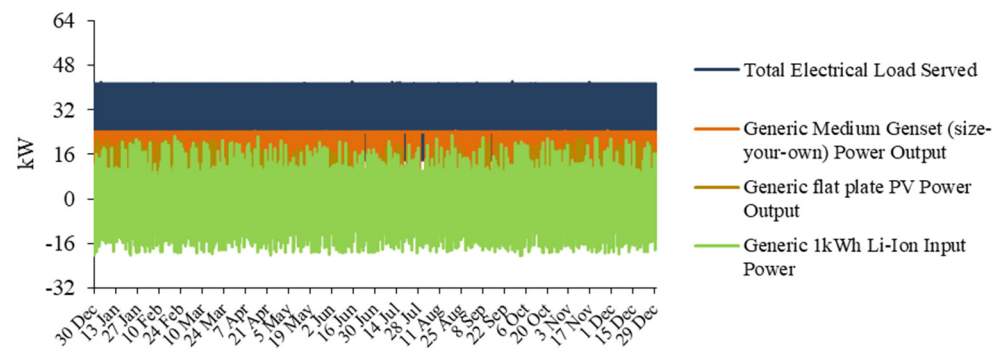


Figure 8. Time series electrical generation.

4.1.2. Thermal Energy Consumption

The facility requires thermal energy consumption of 264 kWh per day, with a peak demand of 20 kWt. Of this peak demand, 11 kWt is allocated for the operation of the vapor absorption refrigeration system (VARS), which yields 7.7 kW of cooling while maintaining a set temperature of 15 °C. Figure 9 illustrates the thermal energy generation available to meet the thermal load, whereas Figure 10 presents the time series of thermal generation. The thermal output produced by the boiler, which serves as the heat recovery system, dominates the overall thermal generation, while the thermal load control mechanism effectively stabilizes the thermal power generation.

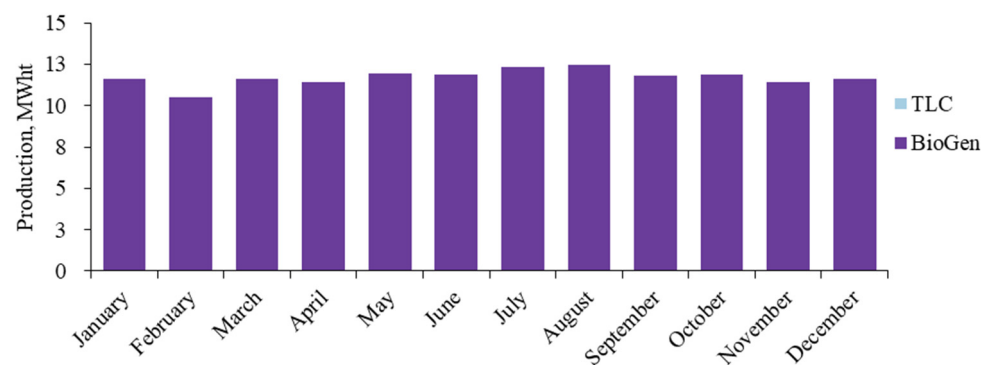


Figure 9. Thermal energy generation.

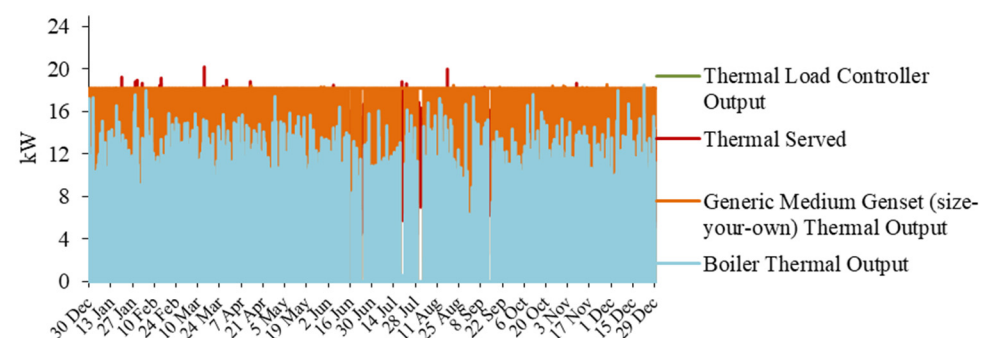


Figure 10. Time series thermal generation.

4.1.3. Solar PV Energy Production

The solar PV system has a nominal capacity of 20.0 kW, with annual electrical energy generation of 35,886 kWh/yr. Table 4 and Figure 11 show the pertinent solar PV parameters and time series electricity generation, respectively. Figure 10 shows solar PV electricity generation is between 6:00 am and 6:00 pm, but the production peaks at noon. In addition, Figure 10 shows that solar production is minimal between July and September,

which coincides with the months of low solar radiation. The LCOE from the solar PV at 0.049 EUR/kWh is far cheaper than the LCOE from the municipal electricity generation at a solar PV penetration of 16.5%.

Table 4. Solar PV generation detail.

Parameter	Unit	Value
Capital cost	EUR	16,645
Specific yield	kWh/kW	1794
PV penetration	%	16.5
Total generation	kW	35,886
Maintenance cost	EUR/yr	189
LCOE	EUR/kWh	0.0469

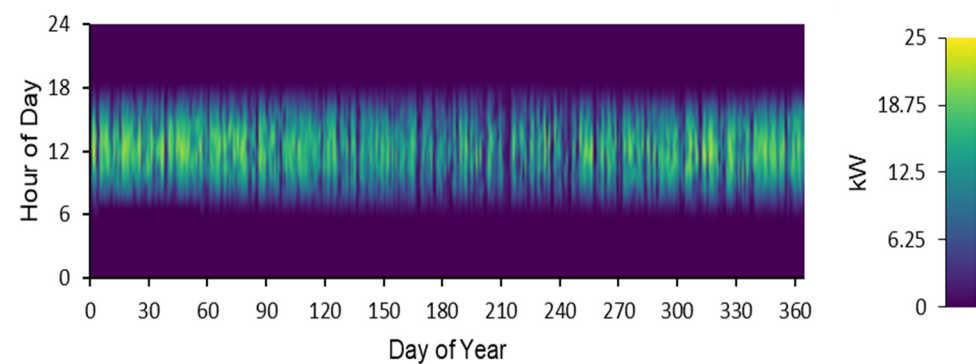


Figure 11. Time series electricity generation by solar PV.

4.1.4. Syngas Generator Energy Generation

The electrical energy output from the syngas generator (CHP), rated at 24.0 kW, is 185,735 kWh/yr, and thermal energy is 140,661 kWhth/yr. Table 5 and Figure 12 show the pertinent parameters of the syngas generator and time series electricity generation, respectively. Figure 12 illustrates that the generator predominantly operates during the early morning and late evening hours (see the yellow patch). This phenomenon can be attributed to the interplay between solar photovoltaic electricity generation and the electricity supply from the battery bank.

Table 5. CHP syngas generator.

Parameter	Unit	Value
Capital cost	EUR	72,000
Fuel consumption	MT/yr	167
Electrical generation	kWhe/yr	185,735
Thermal generation	kWhth/yr	140,661
Maintenance cost	EUR/yr	15,851
Fixed generation cost	EUR/hr	6.04

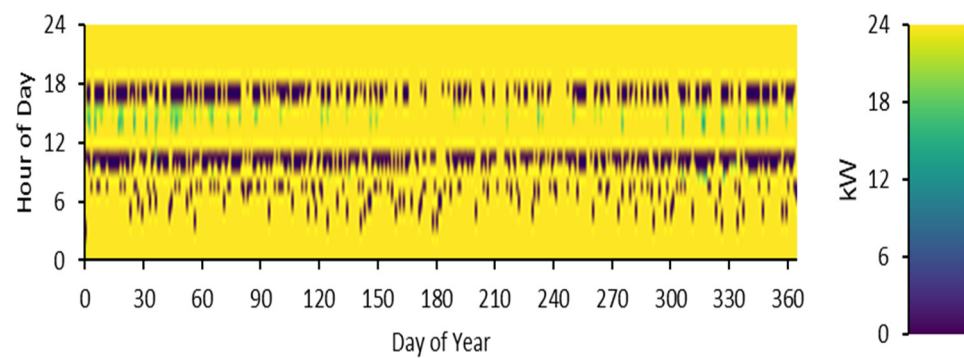


Figure 12. Time series electricity generation by the CHP syngas generator.

4.1.5. Lithium-Ion Battery Storage

The battery energy storage system's nominal capacity is 78.0 kWh. The annual throughput is 23,043 kWh/yr. Table 6 and Figure 13 show the pertinent parameters of the battery storage and the time series operation, respectively. Figure 13 shows that the battery operates at 80% and above charge level between 9:00 a.m. and 5:00 p.m., which is expected due to solar PV performance shown in Figure 11. In addition, the battery charging is below 80% during the months of July and September due to the obvious reason of low solar radiations during these months, as stressed in Section 4.1.3.

Table 6. Lithium-ion battery storage detail.

Parameter	Unit	Value
Rated capacity	kWh	78.0
Maintenance cost	EUR/yr	780
Capital cost	EUR	40,435
Autonomy	h	2.5
Expected life	Yr	10.2
Losses	kWh/yr	2427

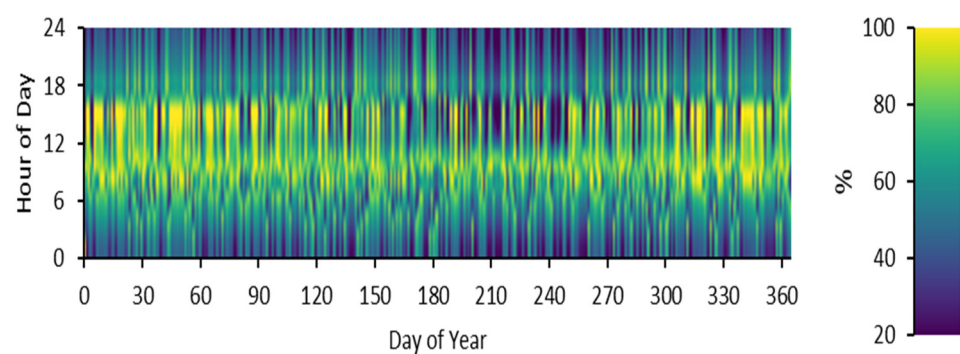


Figure 13. Time series operation of the battery storage.

4.2. Sensitivity Analysis

The optimized system was subjected to some parametric analysis to see the effects on the cost of electricity. The parameters considered are the discount rate, cost of biomass (FC), in EUR/ton, and minimum load ratio. The three parameters were selected to reflect policy, social, and technology dimensions of the sustainability of clean energy interventions. The bounds of the parameters were informed based on experience, historical data, and future technological advancement. Figure 14 shows the response of the levelized cost of energy (LCOE) with the variation in feedstock cost and discount rate. It shows that the LCOE is

increasing with an increase in the discount rate and feedstock cost, which is attributed to both the effect of the time value of money and operational cost. The implication is that a policy that limits interest rates for bioenergy systems and a social solution for feedstock pricing will be favorable to the economic sustainability of the proposed system. One of the possible social solutions to reduce the feedstock cost would be trading the biochar (a carbon sequestration medium) from the plant for the feedstock. Biochar is a carbon-rich material that has gained significant attention in recent years for its potential to improve soil quality and agricultural productivity vis-à-vis improved income of smallholder farmers while also providing environmental benefits, such as carbon sequestration.

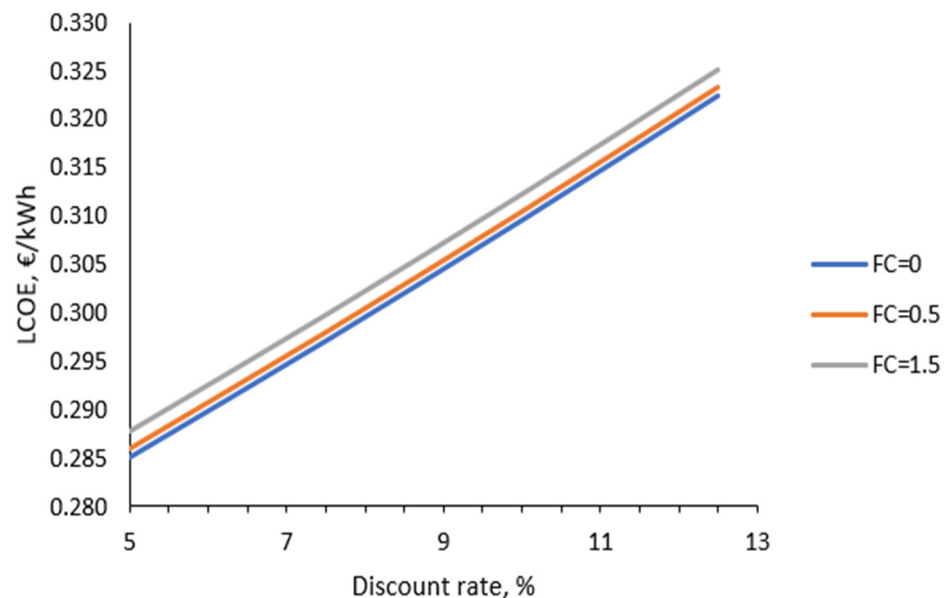


Figure 14. Variation of LCOE with discount rate and feedstock.

Figure 15 illustrates the relationship between the levelized cost of energy, the minimum load ratio, and the discount rate. The data presented indicate that the minimum load ratio exerts a marginal influence on the levelized cost of energy. Nonetheless, it is observed that, on average, a higher minimum load ratio corresponds to a decrease in electricity costs. This reduction can be attributed to enhanced efficiency associated with elevated minimum load ratios.

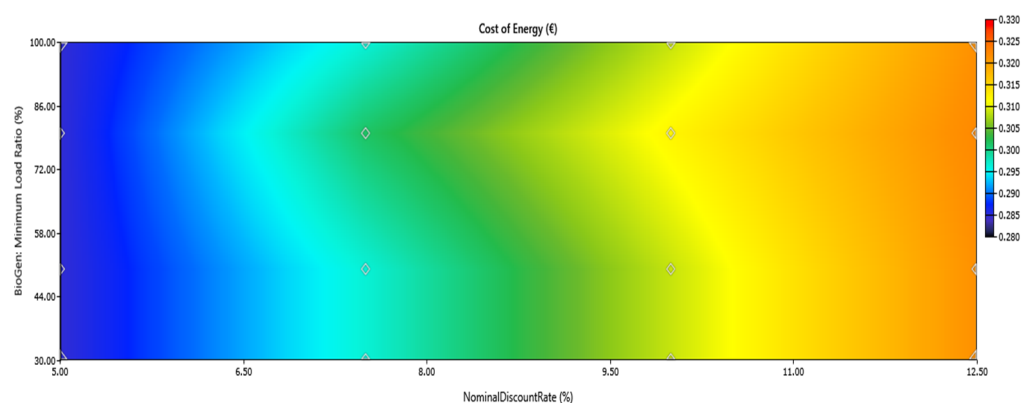


Figure 15. Variation in LCOE with discount rate and minimum load ratio.

5. Conclusions

Access to reliable and affordable energy (SDG7) is critical to global socioeconomic growth. Inadequate access to sustainable energy hampers economic and human development progress, particularly in low- and middle-income nations. Looking at the increase

in electricity access in recent years, it can be seen that Ghana has managed to maintain growth in electricity access of approximately 20% every 10 years for as long as records have been available. Despite Ghana's relatively high energy access rate, power disruptions and energy poverty remain significant issues, especially in rural schools, which hampers quality education [9]. In this regard, the present study proposed a hybrid combined heat and power energy system driven by solar PV and agro-waste (groundnut shells) to meet the energy needs (electricity, heating, and cooling) of Tuna Senior High/Technical School Education Centre (TUSEC) located in Tuna (lat. 9°29'18.28" N and long. 2°25'51.02" W). This system would serve as a model for sub-Saharan African countries like Ghana that align well with energy access challenges. The proposed system's distinctive feature is its capacity to harness the benefits of two renewable energy sources—solar energy and biomass—to produce electricity, heat, and cooling simultaneously. Conventional renewable energy systems traditionally emphasize either electricity or heat generation in isolation. However, in rural educational institutions, there exists a pressing demand for electricity, hot water for hygiene purposes, and cooling for food preservation or comfort, particularly in regions characterized by elevated temperatures like the Sawla-Tuna-Kalba District of Ghana, the location of the plant.

The study utilizes HOMER software to thoroughly analyze technical, economic, and environmental considerations. The system can generate 22.16 MWh of electricity at a unit cost of 0.295 EUR/kWh and produce 110.90 MWh of thermal energy annually. This thermal energy is sufficient to address the heating requirements of the school, in addition to operating a vapor absorption chiller. Environmentally, the system demonstrates sustainability by effectively sequestering a net 0.526 kg of CO₂ emissions yearly. This paper highlights the following:

- (i) The potential for using agro-waste to energy technology combined with solar PV to provide electricity, heat, and cooling, especially in off-grid rural agrarian communities.
- (ii) The potential to improve the quality of education in rural community schools by improving laboratories, technical practice, and long study hours for students.
- (iii) The potential to provide negative carbon emission energy conversion systems.
- (iv) A hybrid energy system with the potential to positively drive SDG 4 (quality education), SDG 7 (affordable and clean energy), and SDG 13 (climate action).
- (v) Techno-economic evidence to support policymakers in formulating public policies that balance energy cost and environmental sustainability.

The proposed plant holds significant policy implications for energy security, sustainability, and socio-economic impact. Notably, promoting financial incentives, such as tax breaks or subsidies, could enhance the plant's attractiveness for both businesses and consumers. Furthermore, policies encouraging farmers to collect and sell biomass residues could provide an additional income stream, bolstering rural economies. Therefore, the next research phase will be to implement and demonstrate the plant in TUSEC to conduct context-sensitive experiments that will map performance using local data. This approach is expected to enhance the adoption and acceptance of the proposed system in Ghana and other sub-Saharan African nations, including Nigeria.

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